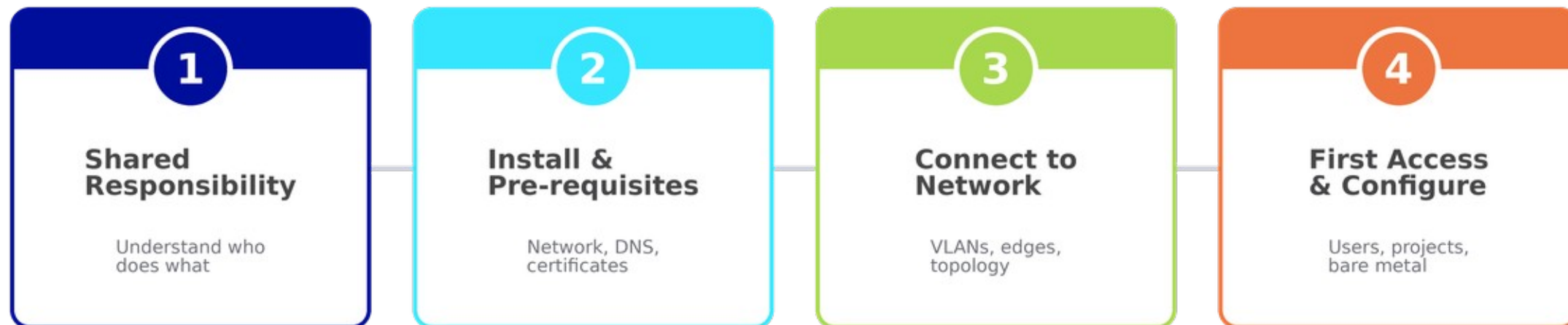


OPCP Journey

The OPCP Journey

Your OPCP journey - 4 first steps

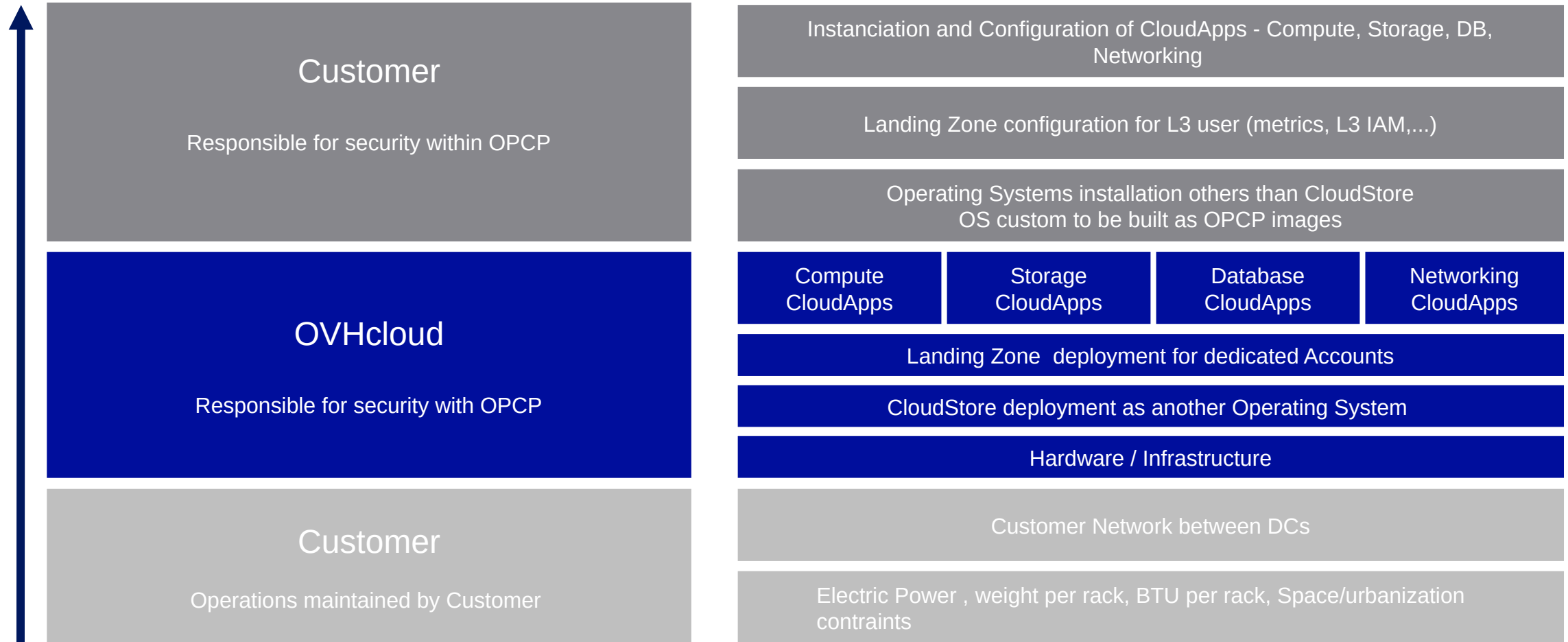


OPCP Shared Responsibility Matrix

Disclaimer : not contractual.

Simplification in the context of a Technical workshop

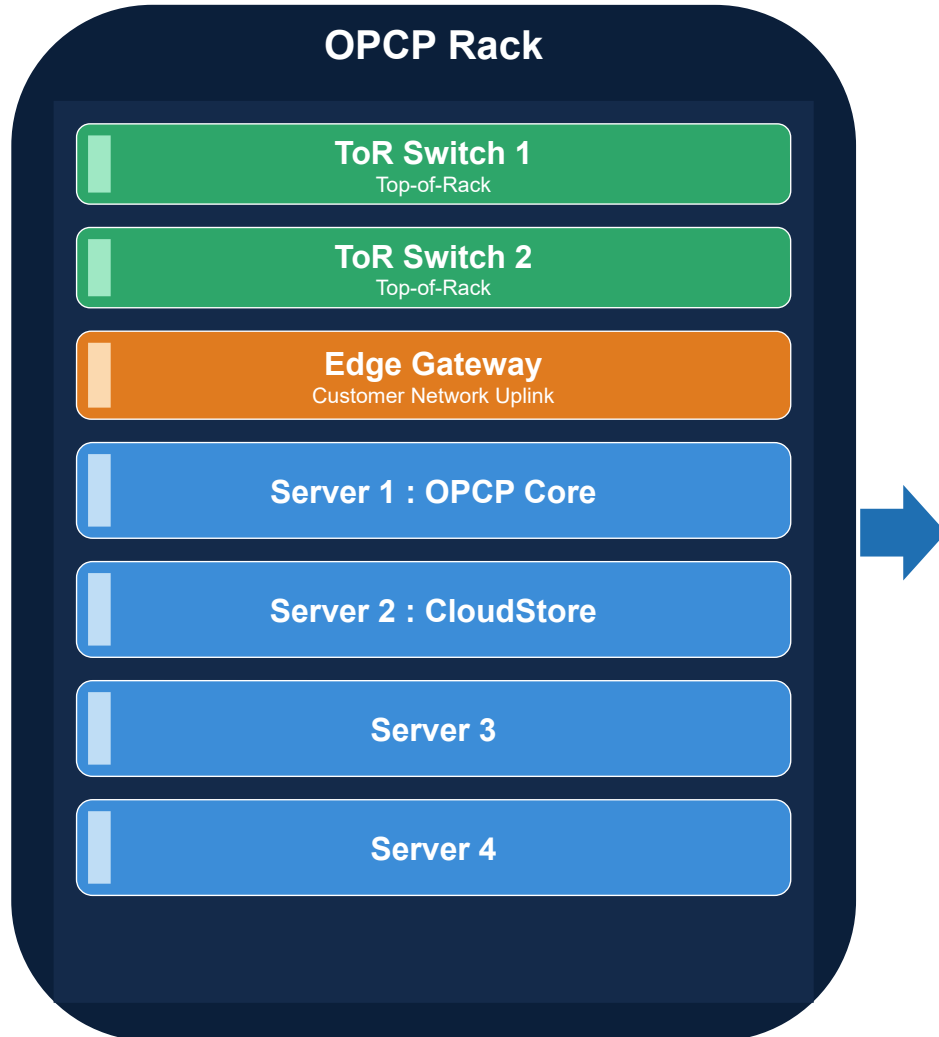
Please refer to precise OVHcloud Legal CGV for OPCP



Installation & Pre-requisites

OPCP Deployment Overview – Rack & Installation Steps

Steps when you receive your OPCP rack



Installation Actions

- 1** **Configure OPCP Core environment + OOB cabling**
Control Plane, Network, servers, DNS, NTP, External logs...
- 2** **Install OPCP Core**
Deploy the OPCP Core platform on 1 or 3 servers
- 3** **Configure the EDGE, cabling OPCP to Customer Net**
For the Data Plane to be accessed. Warning : slot empty !
- 4** **Configure the CloudStore environment (Optional)**
Network, servers, certificates, VLAN...
- 5** **Install CloudStore Application (Optional)**
VCF as an option

OPCP Pre-requisites (OPCP Core)

OVHcloud will handle the configuration of OPCP but the customer provide the elements listed below

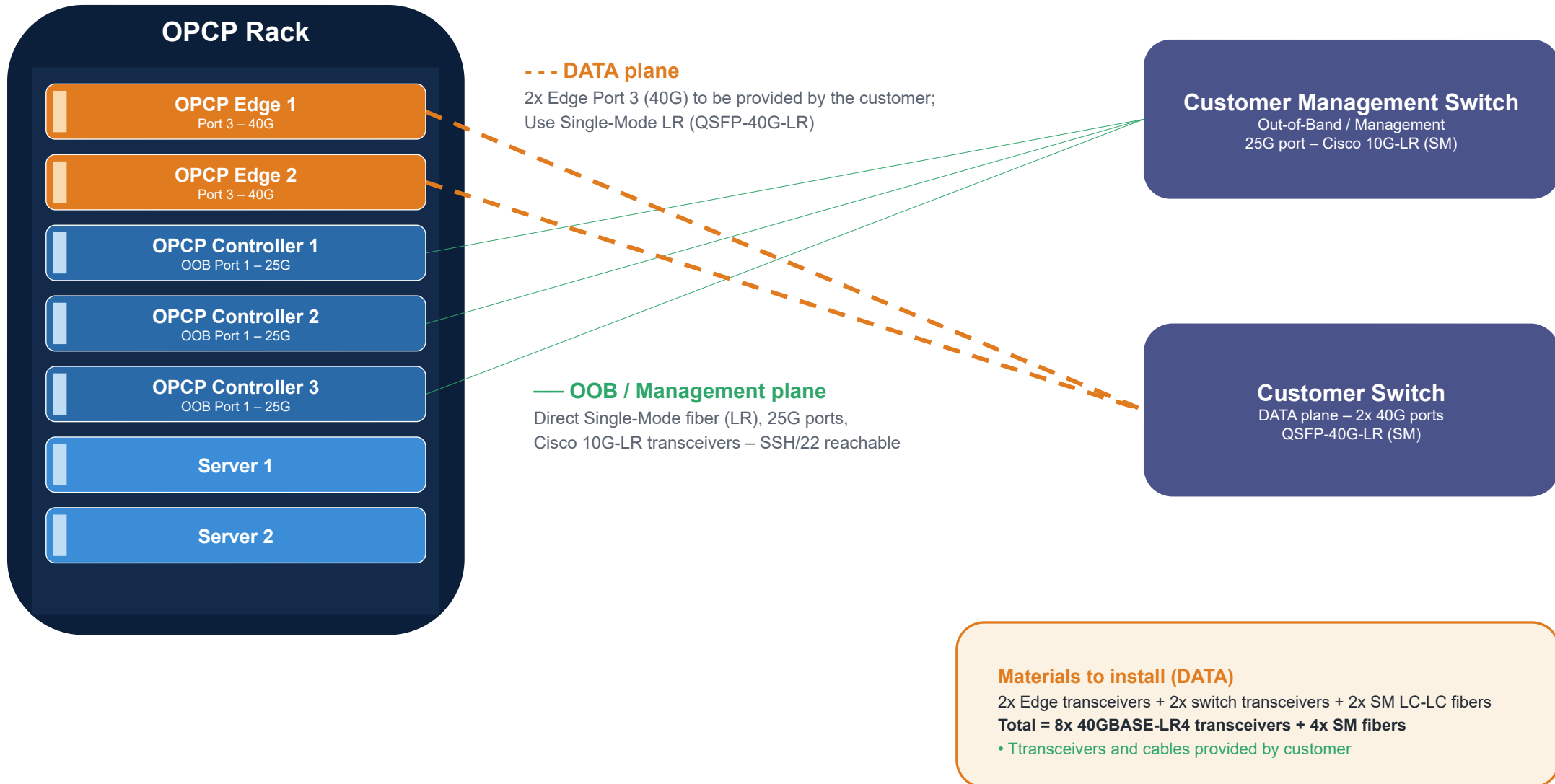
Service	Description
Network	Definition of the management network used by OPCP - Subnet, - VLAN ID, - IP Address, - (VIP address), - Default gateway
Server	Identify the servers that will host OPCP Core - OPCP control plane server name + IP address + VIP
DNS	OPCP need to resolve external domains (for S3 connection, for example) - IP Address
NTP	To able to connect to an external NTP service - IP Address
Certificates	Intermediate certificate (non-root-CA) provided by Customer.
Options	
<i>Backup</i>	<i>To be able to perform backups of the infrastructure - S3 endpoint, - access Key, - private key, - bucket name</i>
<i>LDAP</i>	<i>Centralize user access across all OPCP Keycloak - IP of the AD Servers</i>
<i>SSH keys</i>	<i>Used to give access to opcp controller's after bootstrap. In managed mode provide customer access to: opcp-cli + opcp-diag</i>
<i>Syslog servers</i>	<i>Centralize all the logs from OPCP - IP Address</i>

OPCP Pre-requisites (OPCP Core)

Define customer environment variables for OPCP

Variable	Description
env	• "opcp-paris" (used in many locations : log, backups, etc..)
region	• "IDF" (OPCP region = openstack region)
stage	• "PROD" (alerting, Control plane backup)
base_dns_name	
site	• "PARIS-1" (used in Netbox)
location	• "PARIS" (used in Netbox)

OPCP – Physical Connectivity to Customer Network (Ex.)



Summary : Actions Performed By The Customer

Required	Action Description	OPCP Core	CloudStore (option)
Y	Provide all required information to OVH (via PSMC OVH) Networks, Servers, DNS, NTP, LDAP, Syslog, S3	Cx	Cx
Y	Create Networks, subnets, VLANs (support PSMC OVH)	Cx	Cx
Y	Networks links to connect to the OOB Out Of Band network interfaces	Cx	
Y	Networks links to connect the OPCP rack edge for the Data Plane (via PSMC OVH)	Cx	
Y	DNS zone delegation configured from customer site	Cx	Cx
Optional	Bastion access provided to OVH team if not AirGap	Cx	Cx
Optional	S3 backup bucket provided & reachable from customer network	Cx	Cx

OPCP Pre-requisites (CloudStore) - Optional

OVHcloud will handle the configuration of OPCP but the customer provide the elements listed below

Service	Description
Network configuration	Definition of dedicated network used for CloudStore - Network Name - Subnet Name – CIDR - VLAN ID - Default gateway - DHCP
Server information	Identification of the server that will host CloudStore - Hostname – IP Address – Node UUID - VIP
DNS	OPCP need to resolve external domains (for S3 connection, for example) - DNS Zone Name – Primary DNS Server IP – Secondary DNS Server IP
NTP	To able to connect to an external NTP service – Primary NTP Server IP – Secondary NTP Server IP
Certificates	Intermediate certificate (non-root-CA) provided by Customer.
Options	
<i>Backup Storage S3</i>	<i>To be able to perform backups of the infrastructure - S3 endpoint - access Key - secret key - bucket name</i>
<i>Syslog servers</i>	<i>Centralize all the logs from OPCP - IP Address</i>

FOCUS : HTTPS Certificates

How OPCP obtains its TLS certificates — choose one of four options at configuration time

1

Private CA in the Control Plane

A private Certificate Authority is deployed inside the OPCP Control Plane and issues certificates automatically.

2

Customer-provided private CA

You provide a private CA. OPCP automates validation and deployment of certificates to all endpoints.

3

Customer-provided certificates

You supply ready-made certificates for every endpoint. OPCP installs them as-is.

4

OPCP self-generated (Let's Encrypt)

OPCP automatically generates and renews its own certificates using Let's Encrypt.

Requirements & good to know

- Customer-provided certificates must be trusted by public certification authorities.
- An intermediate (non-root) CA provided by the customer can be used on OPCP.

OPCP Rack

ToR Switch 1

Top-of-Rack

ToR Switch 2

Top-of-Rack

Edge Gateway

Customer Network Uplink

Server 1 : OPCP Core

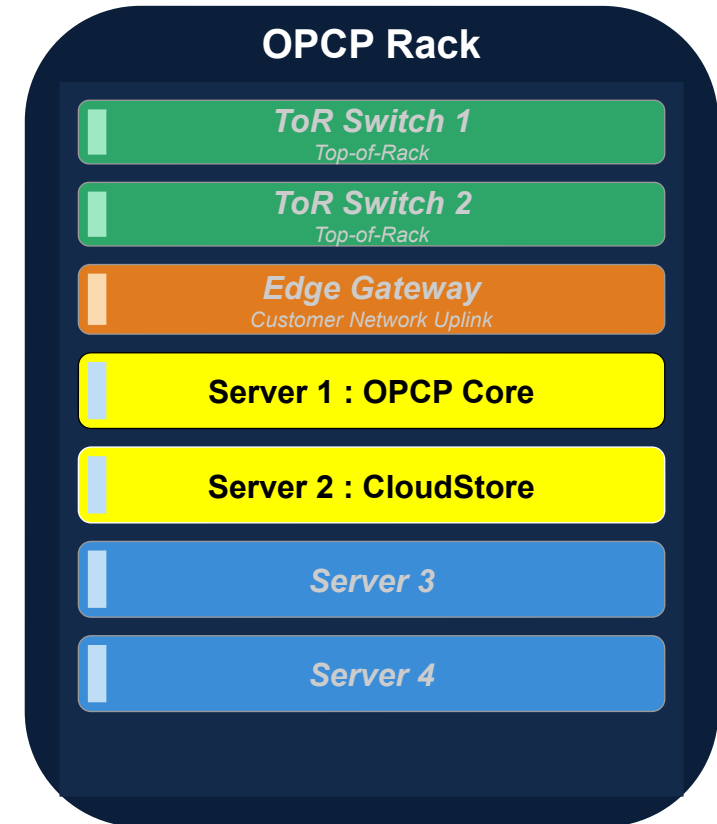
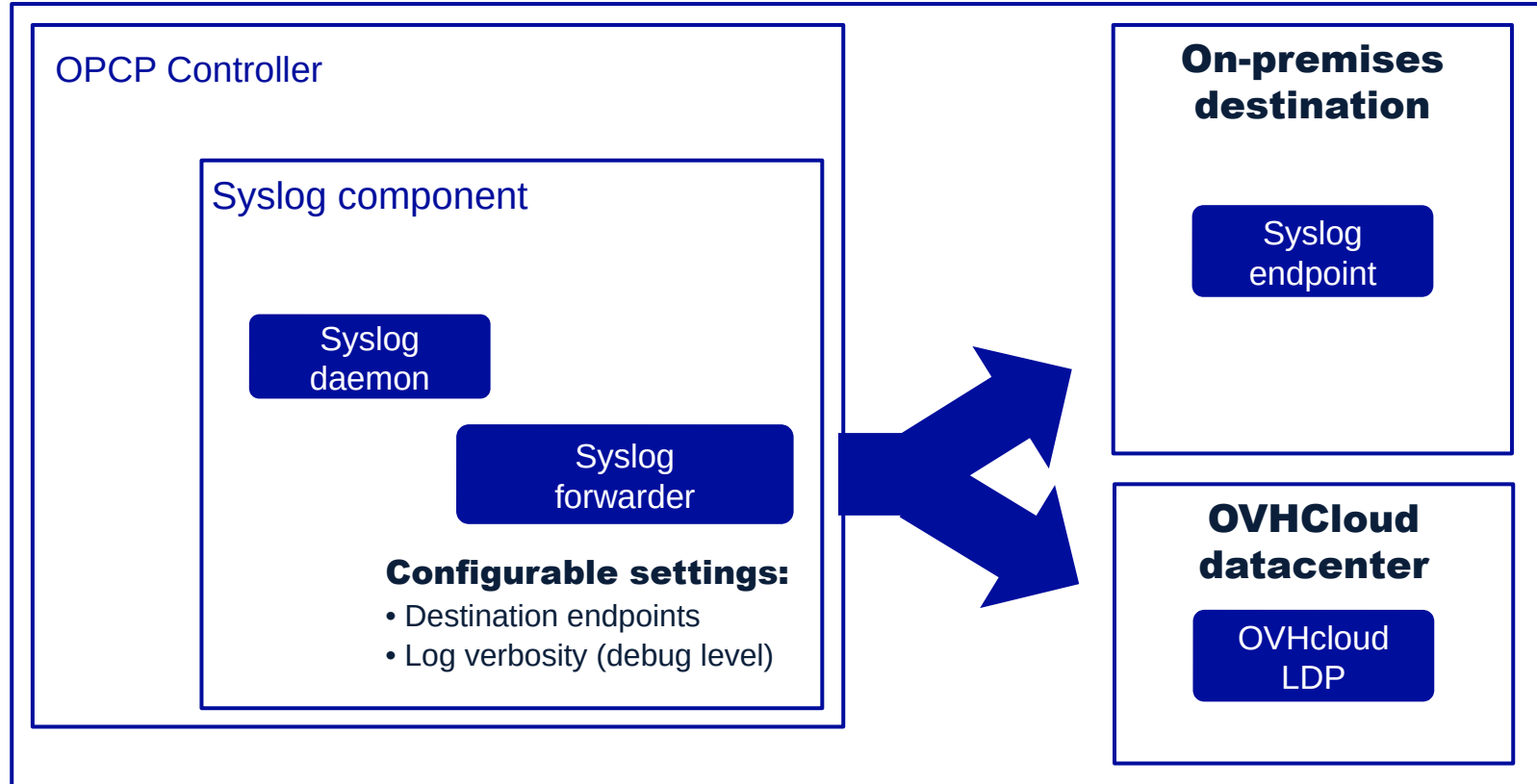
Server 2 : CloudStore

Server 3

Server 4

FOCUS : Externalization Of OPCP Core Logs

Architecture already proven in production for the Bare Metal Pod product

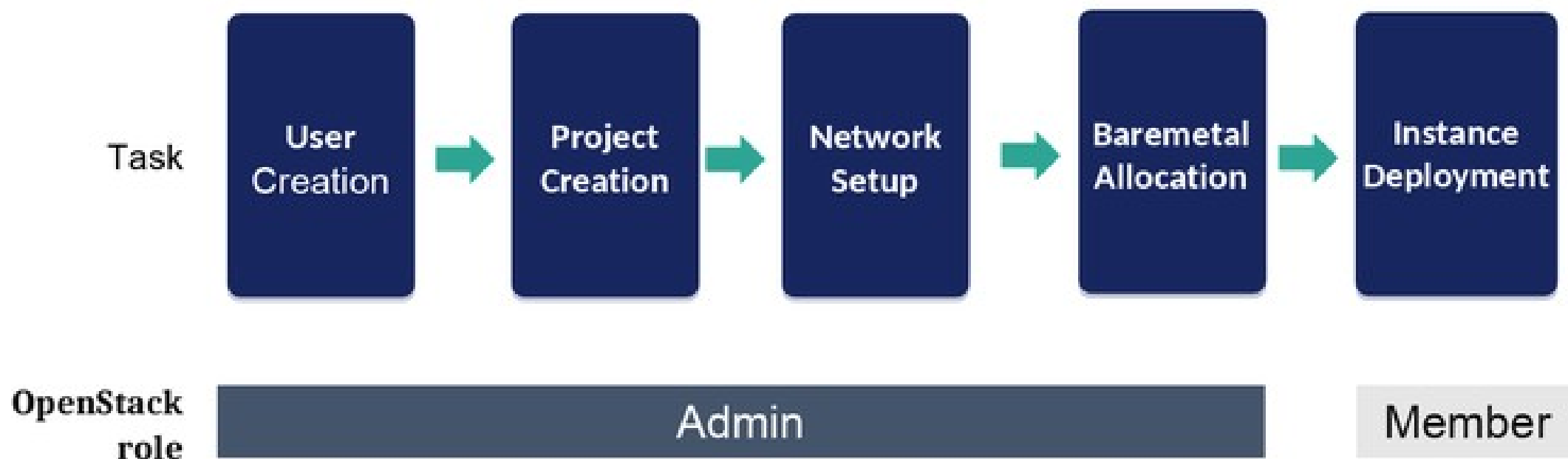


Logs can optionally be forwarded to an on-premises syslog endpoint, to OVHcloud LDP, or to both simultaneously.

First Access

Your OPCP Platform Is Ready ?

You will create users, projects, networks and then deploy your first instance



Web Server Available

<https://admin.keycloak.opcp-croix-demo-3.bmp.ovhgoldorack.ovh/>

Not Secure admin.keycloak.opcp-croix-demo-3.bmp.ovhgoldorack.ovh/realm/

OPCP CORE MASTER

Sign in to your account

Username or email

Password

Sign In

admin.keycloak.opcp-croix-demo-3.bmp.ovhgoldorack.ovh/admin/master/console/#/master/users

KEYCLOAK

master Current realm

Manage realms

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Users

Users are the users in the current realm. [Learn more](#)

User list

Default search Search user → Add user Refresh

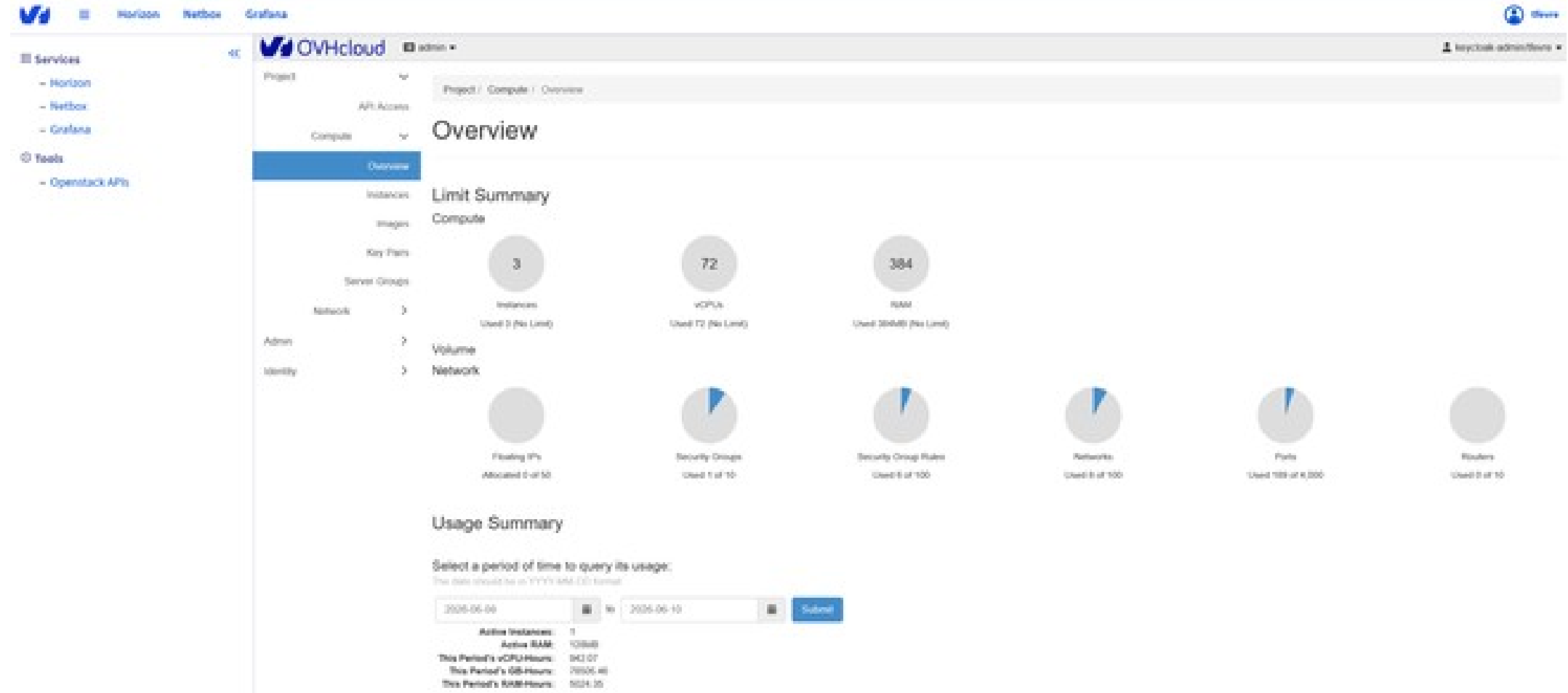
1 - 49

☐	Username	Email	Last name	First na...	
☐	[redacted]	[redacted]@ovh.com	[redacted]	[redacted]	...
☐	[redacted]	[redacted]	[redacted]	[redacted]	...
☐	[redacted]	[redacted]	[redacted]	[redacted]	...
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Web Servers Available

<https://admin.dashboard.opcp-croix-demo-3.com>

OPCP Core Dashboard overview



Step-by-step OPCP Configuration Agenda

As OPCP array has just been delivered, what to do next ?

1

▶ Access, Project and Users

- ▶ Verify API and GUI admin accesses
- ▶ Create management VLAN
- ▶ Create Group and Users
- ▶ Create Projects (isolation)
- ▶ Validate access to Netbox
- ▶ Define the number of Bare Metal servers into the project

2

▶ Bare Metal servers installation

- ▶ Create project Users with reader/member permissions
- ▶ Check BM availability on OPCP
- ▶ Enroll servers into the Project
- ▶ Install instances (to be done by reader/member user)

What About Infrastructure As Code ?

OpenStack provides several ways to manage your infrastructure

Approach	Best for	Learning curve
REST API (cURL)	Debugging, understanding the protocol	High
OpenStack CLI	Day-to-day operations, quick tasks	Low
OpenStack SDK	Custom Python scripts, automation	Medium
Terraform	Infrastructure as Code, reproducible environments	Medium
Ansible	Configuration management, orchestration	Medium

PSMC Labs

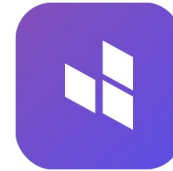
Four Labs to explore, automate and build on OPCP



OPCP First Steps : training content

- ▶ Hands-on introduction to OPCP
- ▶ 1-day training on user management, networking, compute
- ▶ Focus on Keycloak and Openstack CLI and Horizon interface
- ▶ Target audience : customer (SysAdmin) & internal (TAM, support)

Link : <https://opcp-psmc.com:6113>



OPCP Infra As Code : training content

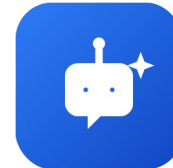
- ▶ Workshop on OPCP automation
- ▶ 1-day training on access methods : API access, OpenStack SDK, Infrastructure-as-Code (Terraform, Tofu, Ansible)
- ▶ Target audience : customer (SysAdmin) & internal (TAM, support)

Beta



OPCP Ai Start Labs : Devops AI

- ▶ Introduction to AI on OPCP environment
- ▶ Experiment Ops Ai Agent + Dev Ai Agent + GPU Sharing + Micro VM on OPCP
- ▶ Target audience : customer (Software development engineers, Machine Learning engineers) for demo purposes
- ▶ Alpha version



OPCP Simulator: Openstack Simulator

- ▶ A pure-Python, in-memory OpenStack simulator (if customer do not have a training environment)
- ▶ Simulate Keystone, Nova, Neutron, Cinder, Ironic, and Glance service interfaces
- ▶ Target audience : customer (SysAdmin) & internal (TAM, support)
- ▶ Alpha version